

Simple Vertebrate Models for Chemical Genetics and Drug Discovery Screens: Lessons From Zebrafish and *Xenopus*

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Chemical genetics uses small molecules to modulate protein function and, in principle, has the potential to perturb any biochemical event in a complex cellular context. The application of chemical genetics to dissect biological processes has become an attractive alternative to mutagenesis screens due to its technical simplicity, inexpensive reagents, and low-startup costs. In vertebrates, only fish and amphibians are amenable to chemical genetic screens. *Xenopus* frogs share a long evolutionary history with mammals and so represent an excellent model to predict human biology. In this review, we discuss the lessons learned from chemical genetic studies carried out in zebrafish and *Xenopus*. We highlight how *Xenopus* can be employed as a convenient first-line animal model at various stages of the drug discovery and development process and comment on how they represent much-needed tools to bridge the gap between traditional in vitro and preclinical mammalian assays in biomedical research and drug development. *Developmental Dynamics* 238:1287–1308, 2009. © 2009 Wiley-Liss, Inc.

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INTRODUCTION

Large-scale genetic screens in model organisms have been used very successfully for many years to identify genes involved in diverse developmental and physiological processes. They have been particularly powerful in identifying key genes and genetic pathways underlying axis specification and pattern formation during embryogenesis in *Drosophila* (Nusslein-Volhard and Wieschaus, 1980), and programmed cell-death in *Caenorhabditis elegans* (Metzstein et al., 1998). However, genetic screens are expen-

sive and time-consuming, particularly if performed in vertebrate model systems. In addition, these screens are of limited use with respect to the genetic dissection of late development and organogenesis. Embryonic development is controlled by a surprisingly small number of genes and signaling pathways, which are redeployed at multiple times and places as development proceeds (Davidson et al., 2002; Van Raay and Vetter, 2004; Reya and Clevers, 2005). Most mutations recovered from genetic screens are not conditional, i.e., they cannot be reversed

at will. Therefore, the mutations will only define the first biological process and time point, where a particular gene is required during embryogenesis. Frequently, the mutation may arrest development and the mutant organism is no longer viable. Hence, it becomes difficult to study late gene functions in organogenesis, physiology, and behavior using a classical genetic approach.

Chemical genetics provides a complementary approach to loss-of-function mutations in the study of complex biological processes (Stockwell, 2000;

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Min et al., 2007). It makes use of small organic molecules with a molecular weight of typically less than 2,000 Dalton (Da) to alter the function of a gene product within a complex cellular context. Chemical genetic strategies are conditional and readily applicable to cells derived from complex organisms such as vertebrates. Compounds can be added or removed at any time, enabling the kinetic analysis of protein functions in vivo. In multicellular organisms, as already stated, chemical genetics offers a complementary approach to loss-of-function mutations or knockdowns with siRNA or morpholino oligonucleotides in the analysis of complex biological processes, such as organogenesis. In addition, chemical genetic screens can also be employed to study the function of maternal gene products, which are not targeted by conventional genetic screens. One approach of chemical genetics, forward chemical genetics, uses the screening of annotated libraries of small organic compounds with experimentally verified biological mechanisms and activities to study biological systems (Stockwell, 2000). This approach circumvents the well-known problems of target identification and lack of mechanistic understanding associated with active compounds recovered from screens using conventional chemical libraries (Root et al., 2003). Forward chemical genetics has, therefore, become increasingly used in cell cultures to identify signaling pathways involved in cellular functions in vitro (Root et al., 2003; Rickardson et al., 2006; Diamandis et al., 2007), and more recently, whole organisms such as *Drosophila*, *C. elegans*, and zebrafish have been used for compound discovery (Min et al., 2007; Tran et al., 2007; Chang et al., 2008). Furthermore, biologically active small molecules recovered from chemical genetics screens also provide important structural information for the development of novel therapeutic agents.

Beyond applications in chemical genetics, small multicellular model organisms offer new opportunities in the drug discovery and drug development pipeline. They can be treated with small molecules in a multi-well format for high-throughput phenotypic analysis to identify novel drug candidates

that are bioactive in a whole organism, where complex cell–cell and cell–matrix interactions remain intact in contrast to in vitro cell-based screening approaches. Furthermore, small model organisms are also beginning to be used at various other stages of the drug discovery process, where they can be useful and cost-effective alternatives to mammalian models. These uses include drug target identification and validation, lead compound characterization and optimization, and the assessment of drug toxicity. Simple and cost-effective maintenance together with abundant experimental techniques and molecular tools have made zebrafish the only vertebrate model used to date for chemical genetics and large-scale in vivo drug screens (Zon and Peterson, 2005). Amphibians offer many of the same experimental advantages that have favored the use of zebrafish in the past, such as rapid extrauterine development, the transparency of developing tadpoles, and the permeability of the skin for small molecules, but they have until recently not been employed for large-scale chemical screening. In this review, we will introduce *Xenopus* embryos and tadpoles as alternative vertebrate model organisms for chemical genetics and whole-organism-based drug discovery screens. Furthermore, we show that *Xenopus* represents a cheap and efficient in vivo bioassay tool that can contribute to several aspects of the drug development process, including drug target identification, lead compound discovery, and estimation of drug toxicity.

ANIMAL MODELS FOR ORGANISM-BASED CHEMICAL SCREENING

Organism-based chemical screening tests the ability of each compound of a library to induce a specific phenotype in an organism. Such screens are, therefore, analogous to classic forward-genetic screens in model organisms. There are two purposes for carrying out chemical screens in model organisms. One is to promote research in a given area by obtaining small molecules, such as antagonists, that can be used as conditional research tools to investigate fundamental questions in development, physiology, and

behavior. The second and increasingly more important application of these screens is to identify drug candidates that can be potentially used for therapeutic purposes. An important advantage of organism-based chemical screening is the fact that compounds are tested in the context of an intact organism rather than under artificial in vitro conditions. Many diseases affect organs as a whole, and most organs cannot be reconstituted in vitro. Small molecules discovered by virtue of their ability to induce a specific, desirable phenotype in a whole organism are likely to fulfill efficacy and specificity requirements that ultimately need to be met by promising drug candidates entering clinical development. They have to be cell-permeable, devoid of obvious toxicities, effective, and possess favorable, pharmacodynamic and pharmacokinetic profiles. Drug discovery in the whole organism, therefore, combines screening and animal testing in one step.

Biomedical research exploits a range of model organisms to gain insights into fundamental biological processes and disease mechanisms. The nematode *C. elegans* and the fruit fly *Drosophila melanogaster* are popular genetic model organisms and they have recently also been used as screening tools in drug discovery (Segalat, 2007). An important limitation of invertebrate models is, however, the fact that they lack many directly extrapolatable complex organs, such as a cardiovascular system, an immune system, and kidneys, relevant to human biology and physiology. Vertebrate models, by contrast, usually have all the tissues afflicted by common human disease. However, not all vertebrate animal models are equally suitable for organism-based drug discovery screens. Table 1 compiles some of the general advantages and disadvantages of the commonly used multicellular model organisms. Animal models to be employed for organism-based chemical screens have to be small, low-cost, and compatible with simple culture conditions to be suitable for the technologies of modern high-throughput screening. Therefore, organisms producing large numbers of embryos are essential, which means that chicken and mouse are not suitable for chemical screens. The or-

ganisms also have to allow rapid penetration of small molecules. *Drosophila* and *C. elegans* are surrounded by a thick cuticle that is a physical barrier to the penetration of small molecules limiting the access to tissues and organs in these organisms.

Among the vertebrate models only zebrafish and *Xenopus* fulfill the above-mentioned criteria for high-throughput organism-based chemical screening. The animals have simple husbandry requirements, are fecund, and generate large clutches of eggs. Fertilization and embryonic development are external. The resulting embryos and larvae are small enough to grow in microformat screening plates. *Xenopus* scores well in most of the categories. Their eggs can be easily obtained in numbers of several thousands at any time during the year by simple hormone injection and they can then be synchronously fertilized in vitro. This facilitates biochemical, pharmacological, and statistical analyses. They develop in simple salt solutions at room temperature. Compounds can be added to the bathing media and the vitelline membrane around the embryo is highly porous and thus accessibility of compounds to the embryo is assumed to be good. Later, drugs and small molecules can be rapidly absorbed through the skin and gills, though this has not been studied in detail. *Xenopus laevis* embryos are bigger than zebrafish, but they can still be screened in 48- or 96-well plates (Brändli, 2004; Tomlinson et al., 2005). Embryos of *Xenopus tropicalis*, the diploid sister species of *Xenopus laevis*, are half the size of *Xenopus laevis* embryos and are, therefore, more suitable for assaying in 96-well plates. Overall, zebrafish and *Xenopus* share similar experimental advantages with regard to their utility for high-throughput chemical screening.

ADVANTAGES AND DISADVANTAGES OF *XENOPUS* AS A VERTEBRATE MODEL ORGANISM FOR CHEMICAL SCREENING

Zebrafish and *Xenopus* are both undoubtedly well suited to whole-organ-

ism based small-molecule screens because large numbers of embryos can be arrayed in multiwell plates, along with compounds from a chemical library. The first large-scale chemical library screen using a vertebrate model was reported by Peterson et al. (2000) who screened 1,100 compounds against zebrafish embryos arrayed in 96-well plates to identify molecules that modulate embryogenesis. The use of *Xenopus* as an alternative vertebrate model for chemical screens has been proposed in the past (Brändli, 2004; Tomlinson et al., 2005), but until recently zebrafish has remained the only vertebrate model used for large-scale chemical screens (Zon and Peterson, 2005). This has been in part due to the availability of chemically induced zebrafish mutants that model human disease (Shin and Fishman, 2002; Rubinstein, 2003). The versatility of *Xenopus* must, therefore, be compared to the well-established properties of the zebrafish system. The recent report of genetic screens performed in *Xenopus tropicalis* (Goda et al., 2006) indicate that *Xenopus* mutants that phenocopy human disorders may be recovered in the future. The *X. tropicalis* genome is now nearly complete (JGI *X. tropicalis* genome assembly 4.1; <http://genome.jgi-psf.org/Xentr4/>), public databases harbor the nucleotide sequences of almost 2 million expressed sequence tag (EST) from *X. tropicalis* and *X. laevis* (dbEST; <http://www.ncbi.nlm.nih.gov/dbEST/>), and DNA microarrays are available for both *Xenopus* species. Thus, *Xenopus* is rapidly becoming a model organism armed with an impressive collection of genomic and transcriptomic tools.

The closer a model organism is in evolutionary terms to the target organism, i.e., humans, the more reliably results from studies in the model organism translate to humans. Figure 1 shows a phylogenetic tree highlighting the evolutionary distances between various animal model organisms used in biomedical research. *Xenopus* has a common evolutionary history with mammals that is an estimated 90–100 million years longer than between zebrafish and mammals. Since both are tetrapods, *Xenopus* and mammals share extensive synteny at the level of the genomes

and have many similarities in organ development, anatomy, and physiology (Christensen et al., 2008; Raciti et al., 2008). A high degree of sequence conservation and structure between the proteins of model species and humans is desirable, since this will increase the likelihood that bioactive compounds recovered in chemical screens in model organisms will retain activity and specificity for preclinical studies in mice and eventually for clinical studies in humans. As stated, *Xenopus* as a tetrapod is evolutionarily closer to humans than zebrafish. There are also significant differences at the genomic level. Genome and EST sequence data support the notion that the *X. tropicalis* genome is diploid in nature (Hellsten et al., 2007). In contrast, the teleost lineage of fish underwent a genome duplication event after diverging from tetrapods about 450 million years ago (Postlethwait et al., 1998; Postlethwait, 2007). Zebrafish, therefore, possess two copies of many mammalian genes. Such redundancy can confound extrapolation of the effect of single gene knockdown or mutation from fish to mammals and complicate the generation of zebrafish models of congenital human disease. In addition, comparative genomics has identified entire gene families, such as the KRAB domain containing C2H2 zinc finger (KRAB-ZF) transcription factors (Emerson and Thomas, 2009) that first arose in tetrapod vertebrates and are absent in the zebrafish genomes.

There are drawbacks that apply to both animal models. Most advantages of zebrafish and *Xenopus* as a model for chemical screens are limited to embryonic and larval stages and do not apply to adult organisms, since they are typically too large to be employed in whole-organism-based screens. Despite these limitations, zebrafish larvae and *Xenopus* tadpoles contain most organs and tissues affected by common human diseases, including a cardiovascular system, a digestive tract, excretory organs, sensory organs, a hematopoietic system, and a central nervous system. Being tetrapods, *Xenopus* tadpoles will develop lungs and limbs, which are absent from zebrafish. Importantly, the organs of *Xenopus* tadpoles are morphologically and functionally more similar to their human counterpart than those

TABLE 1. Advantages and Disadvantages of Common Animal Models for Organism-based Chemical Screening

	<i>C. elegans</i>	<i>Drosophila</i>	Zebrafish	<i>Xenopus</i>	Chicken	Mouse
Broodsize	250-350	80-100	100-200	1000-5000	1	5-8
Costs per embryo	low	low	low	low	medium	high
High-throughput multiwell-format screening	good	good	good	good	poor	poor
Access to embryos	good	good	good	good	good	poor
Micromanipulation of embryos	limited	limited	fair	good	good	limited
Genome	known	known	known	known*	known	known
Genetics	good	good	good	fair*	none	good
Knock downs (RNAi, morpholinos)	good	good	good	good	limited	limited
Transgenesis	good	good	good	good	poor	good
Evolutionary distance to man	very distant	very distant	distant	intermediate	intermediate	close

**Xenopus tropicalis*.

Color code: green, best in category; red, worst in category.

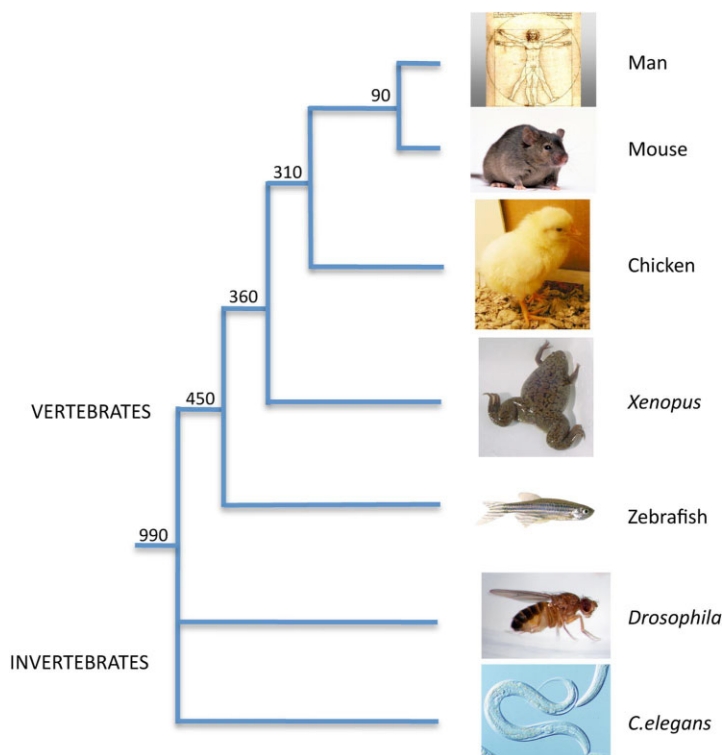


Fig. 1. Phylogenetic tree showing the main animal models commonly used in biomedical research and their evolutionary relationship. The divergence times in millions of years ago (Mya) are shown based on multigene and multiprotein studies (Hedges and Kumar, 2002; Hedges, 2002; Ureta-Vidal et al., 2003). Branch lengths are not proportional to time.

found in zebrafish, as we will illustrate with three examples: the heart, immune system, and the kidneys. The zebrafish heart contains two chambers, a single atrium and a single ventricle, with endocardial cushions forming an atrioventricular valve (Stainier, 2001). In contrast to amphibians and mammals, the zebrafish heart does not progress to septation. The heart of *Xenopus* tadpoles is an evolutionary intermediate between the two-chambered heart of fish and the four-chambered heart of mammals (Mohun et al., 2000; Warkman and Krieg, 2007). It consists of a single ventricle, and left and right atrial chambers separated by a septum. In general, the events of *Xenopus* heart development closely resemble the equivalent processes of heart development in mammals including a leftward bend of the outflow tract, the presence of a spiral valve in the outflow tract and an atrioventricular valve to separate the atria and ventricle, asymmetric division of the atria early in development (with the right side being larger), and the presence of trabeculae within the thickened wall of the ventricular myocardium (Mohun et al., 2000; Warkman

and Krieg, 2007). Differences are also evident with the immune system. While the antibody-based, adaptive immune system is present in all jawed vertebrates (gnathostomes) (Klein and Niko-laidis, 2005), zebrafish and *Xenopus* differ considerably with regard to the presence of major lymphoid tissues, immunoglobulin gene organization, and the ability to perform immunoglobulin class switching (Du Pasquier et al., 1989; Traver et al., 2003). Bone marrow, gut-associated lymphoid nodules, and primitive lymph nodes are absent from zebrafish but present in *Xenopus*. There are also significant differences in the gene organization, gene usage, and gene number of immunoglobulin genes, whereby *Xenopus* shares many features with mammals (Du Pasquier, 2001). Zebrafish express a new immunoglobulin heavy chain subtype, immunoglobulin Z, which does not have a counterpart in mammals (Danilova et al., 2005). Furthermore, isotype class switching is a process that has its earliest evolutionary roots in amphibians and does not occur in fishes (Traver et al., 2003). In summary, *Xenopus* display an efficient immune system that is very similar to mammals and includes rearranging T-cell receptors (TCR) and immunoglobulin genes, as well as major histocompatibility complex (MHC) class I- and class II-restricted T cell recognition. With regard to the excretory system, the pronephric kidneys of zebrafish and *Xenopus* kidneys differ in their global structure from the adult mammalian kidneys, but fulfill the same essential physiological functions in solute reabsorption, water and ion homeostasis, pH regulation, and waste product excretion. This is underscored at the molecular level by a recent large-scale gene expression study that revealed a remarkable conservation between the nephron organization of *Xenopus* tadpole kidneys and adult mammalian kidney (Raciti et al., 2008). Overall, these traits favor the use of amphibians for large-scale in vivo drug screens and suggest that they may unveil mechanisms and pathways relevant to human disease and therapy. There is no doubt that mammals, such as mice and rats, clearly reflect human physiology better. However, there are no other vertebrates of the tetrapod superclass that offer the same advantages of *Xenopus*. Their free-living embryos are amenable to

large-scale, high-throughput chemical screens. Therefore, *Xenopus* provides the optimal trade-off between experimental use, cost-efficient performance, and biological relevance to humans.

LESSONS FROM PHENOTYPE-BASED CHEMICAL LIBRARY SCREENS IN ZEBRAFISH

Assessing the Feasibility of Phenotypic Chemical Library Screens in Zebrafish

Compounds of chemical libraries are usually stored as stock solutions in dimethyl sulfoxide (DMSO), which also serves as a vehicle to improve solubility of the compounds in the aqueous solutions used to culture embryos and facilitates compound permeation into cells. Both zebrafish and *Xenopus* embryos are surprisingly tolerant to a range of DMSO concentrations (Brändli, 2004; Chan and Serluca, 2004). A final concentration of 1% DMSO is compatible with normal development in both species, which indicates that compounds can be screened at a wide range of different compound concentrations in larvae and tadpoles. Embryos from both species are small enough (about 1 mm diameter) to be arrayed in 48- or 96-well plates and the analysis for responses to chemical treatment can be performed by microscopy, in situ hybridization, or reporter readout. Despite these shared advantages, whole-organism chemical screens had until recently not been carried out in *Xenopus* and zebrafish have led the way (Table 2).

The potential of zebrafish to be used as a vertebrate model for chemical screens was first assessed by Peterson and colleagues (Peterson et al., 2000). They used wild-type zebrafish embryos to screen for chemical modifiers of development using a strategy analogous to genetic screens, where mutations are identified that affect development of different organ systems. In their screen, 1,100 synthetic small molecules were randomly chosen from the DIVERSet library (Chembridge Corp.) and added to the embryos. The effects of the individual compounds on the embryo's morphology and physiology were examined visually under the

dissection microscope. From this screen, 2% of the compounds cause lethality, and 1% caused specific phenotypes leading to the identification of compounds that affect development of the central nervous system, the cardiovascular system, pigmentation, and the ear. Several of the identified compounds were highly potent and specific by acting in the nanomolar range and frequently affecting only one organ. For example, the molecule 32P6, subsequently renamed concentramide, affects heart chamber patterning by causing the ventricle to form in the atrium (Peterson et al., 2000, 2001). Concentramide acts with an EC₅₀ of 2 nM and does not appear to cause additional side effects (Peterson et al., 2001). Peterson and colleagues also took advantage of the fact that active compounds can be administered at different times during zebrafish development, which allowed them to determine the exact time point when a target protein's activity was required during development. This provides temporal insights into development, which frequently cannot be obtained using conventional genetic mutants, as the mutant proteins may not permit survival beyond early embryogenesis. While the precise targets of the active molecules recovered in the in vivo chemical screen remain unknown to date, the work by Peterson and colleagues demonstrated that the zebrafish embryo is permeable to many small molecules underscoring the utility of zebrafish as a high-throughput in vivo bioassay system. Importantly, it was subsequently demonstrated that most of the compounds identified by Peterson et al. (2001) gave comparable results when applied to *Xenopus* embryos (Tomlinson et al., 2005).

The Spectrum of Chemical Screens Performed in Zebrafish

Over the last few years, phenotype-based developmental screens using wild-type zebrafish embryos have identified multiple compounds that cause general defects in embryogenesis and now await further characterization (Sternson et al., 2001; Moon et al., 2002; Spring et al., 2002; Wong et al., 2004). Chemical screens were also

TABLE 2. Summary of Chemical Library Screens Performed in Zebrafish and Xenopus

Library name	Source	Compounds screened	Concentration	Organism (Genotype)	Phenotypic readout	Reference
DIVERSet E	Chembridge	1,100	10 µg/ml	Zebrafish (wild type)	Central nervous system, ear, cardiovascular system, pigmentation	Peterson et al., 2000
1,3-dioxane library	Schreiber laboratory	1,300	60 µM	Zebrafish (wild type)	Embryogenesis	Sternson et al., 2001
Compounds with biaryl containing medium rings	Schreiber laboratory	1,412	10 µM	Zebrafish (wild type)	Embryogenesis	Spring et al., 2002
Trisubstituted triazines	Chang laboratory	109	10 µM	Zebrafish (wild type)	Embryogenesis	Moon et al., 2002
Tagged trisubstituted triazines	Chang laboratory	1,536	50 µM	Zebrafish (wild type)	Brain, Eye	Khersonsky et al., 2003
Bioactive small molecules	Various suppliers	100	1,10 & 100 µg/ml	Zebrafish (wild type)	Heart rate	Milan et al., 2003
DIVERSet E	Chembridge	5,000	2 µg/ml	Zebrafish (wild type)	Erythropoiesis	Shafizadeh et al., 2004
DIVERSet E	Chembridge	5,000 & 7,000	2 µg/ml	Zebrafish (<i>grl</i> mutant)	Suppression of vascular defect	Peterson et al., 2004; Hong et al., 2006
Expanded 1,3 dioxane library	Schreiber laboratory	384	5-10 µM	Zebrafish (wild type)	Embryogenesis, organogenesis	Wong et al., 2004
Tagged trisubstituted triazines	Chang laboratory	1,536	10 µM	Zebrafish (wild type)	Pigmentation	Jung et al., 2005
DIVERSet E	Chembridge	16,320	5 µg/ml (20 µM)	Zebrafish (<i>crb</i> mutant)	Suppression of mitotic defects	Stern et al., 2005
DIVERSet E	Chembridge	16,320	20 µM	Zebrafish (wild type)	Suppression of mitotic defects	Murphey et al., 2006
a) NINDS Custom Collection	NIH/NINDS	1,040	5 µg/ml (20 µM)	Zebrafish (wild type)	Haematopoiesis	North et al., 2007
b) SpecPlus Collection	MicroSource	960				
c) ICCB Known Bioactives	BIOMOL	480				
Spectrum Collection	MicroSource	2,000	25 µM	Zebrafish (caudal fin amputation)	Inhibition of regeneration	Mathew et al., 2007
LOPAC1280	Sigma Aldrich	1,280	30 µM	Zebrafish (transgenic VEGFR2: GRCFP)	Angiogenesis	Tran et al., 2007
a) DIVERSet E Collection	Chembridge	5,580	10 µM	Zebrafish (wild type)	Dorsalization	Yu et al., 2008b
b) Spectrum Collection	MicroSource	1,840				
c) Bioactive Compounds	Sigma-Aldrich	150				
DIVERSet E	Chembridge	5,760	5 µg/ml (20 µM)	Zebrafish (wild type)	Organogenesis	Sachidanandan et al., 2008
a) Diversity set	NCI	1,990	20 & 40 µM	<i>Xenopus laevis</i> (wild type)	Pigmentation	Tomlinson et al., 2009a; Tomlinson et al., 2009b
b) Gen-Plus collection	MicroSource	960				
LOPAC1280	Sigma Aldrich	1,280	20 µM	<i>Xenopus laevis</i> (wild type)	Blood vasculature, lymphatics	Kälin et al., 2009

designed to identify novel bioactive compounds that affect the development of specific organs, physiological activities, or cellular processes in zebrafish (Table 2). These include screens to identify compounds affecting dorsoventral patterning (Yu et al., 2008b), hematopoiesis (North et al., 2007), erythropoiesis (Shafizadeh et al., 2004), vasculature (Tran et al., 2007), pigmentation (Jung et al., 2005; North et al., 2007), fin regeneration (Mathew et al., 2007), and morphogenesis of the brain, eyes, and other tissues and organs easily identified by visual inspection of compound-treated embryos (Khersonsky et al., 2003; Sachidanandan et al., 2008). Finally, screens to identify modifiers or suppressors of zebrafish mutant phenotypes have been developed (Peterson et al., 2004; Stern et al., 2005; Hong et al., 2006). We will now review the specifics of these screens in greater detail to highlight key lessons that can be learned from the careful analysis of the published literature. Technical aspects of performing chemical screens in zebrafish have been reviewed elsewhere in detail (Chan and Serluca, 2004; Peterson et al., 2004; Murphey and Zon, 2006; Berger and Currie, 2007).

Exploring the Advantages of In Vivo Screens Over Cell-Based Assays

One of the great hopes of phenotypic whole-organism-based drug screens is that they may lead to the identification of novel bioactive compounds that could not be recovered using standard cell culture-based phenotypic screens. In contrast to cell cultures, cells in whole organisms are non-transformed and found in their normal context of three-dimensional cell–cell and cell–matrix interactions, ingested compounds may become active as metabolites, and compounds exerting specific biological effects indirectly by acting on adjacent tissue may be recovered.

As an alternative approach to standard cell culture-based drug screens, a chemical screen in zebrafish embryos was devised to identify lead compounds that have effects on the cell cycle and hence could act as novel chemotherapeutics for cancer treat-

ment (Murphey et al., 2006). The zebrafish-based screen was performed with a chemical library that was previously screened for mitotic inhibitors with cell lines (Haggarty et al., 2000). Of the 29 identified compounds affecting cell cycle activity in vivo, 14 compounds were novel. Seven compounds were also active in mammalian and/or zebrafish cell cultures. The remaining seven compounds were only active in the context of the intact organism. Whether these compounds also elicit in vivo cell cycle activity in mammalian animal models has not been tested to date. Interestingly, three compounds were inactive in vitro due to the presence of serum. The serum-free culture conditions for zebrafish embryos, therefore, enabled the identification of biologically active compounds that would absorb to serum proteins and score as inactive in the cell culture screens. Taken together, the study demonstrates that whole organism-based screens can indeed identify compounds with novel activities, which had been missed in cell culture-based phenotypic. Cell-based and organism-based chemical screens, therefore, complement each other in identifying novel lead compounds.

Overcoming the Limits of Visual Screens in Wild-Type Zebrafish Embryos

Despite the high degree of transparency during embryogenesis, not all tissues of the zebrafish embryo are easily identified and monitored by light microscopy. Furthermore, as development proceeds, pigmentation will hamper phenotype detection. This problem can be overcome by supplementing the growth medium with the tyrosinase inhibitor 1-phenyl-2-thiourea (PTU), which is routinely used to inhibit pigmentation production and is usually well tolerated by zebrafish embryos and larvae (Peterson et al., 2000; Peterson and Fishman, 2004). The recent generation of *casper* (White et al., 2008), a robust, fertile, and easily maintained transparent zebrafish mutant line that lacks all melanocytes and iridophores in embryos, larvae, and adulthood, may in future replace the use of PTU in chemical screens. In addition, the development of transgenic zebrafish expressing re-

porter genes under the control of tissue-specific promoters significantly expands the possibilities of phenotype detection in wild-type zebrafish embryos subjected to chemical screening. Burns and colleagues reported a first assay for small molecules that modulate the heart rate in transgenic zebrafish embryos (Burns et al., 2005). They developed an automated 96-multiwell plate assay for heart rate using automated fluorescence microscopy of transgenic embryos expressing green fluorescent protein (GFP) in the myocardium. The assay can be used to rapidly assess the differing pharmacokinetic profiles for small molecules that influence the heart rate, and is suitable for high-throughput chemical screening. As a note of caution, it should be mentioned that the practical use of zebrafish as a model to assess cardioactive drug candidates was recently questioned because of the high concentrations that must be reached to see the pharmacological effects (Mittelstadt et al., 2008). Transgenic zebrafish expressing GFP in the embryonic vasculature were used for the first time to screen chemical libraries for compounds eliciting anti-angiogenic effects in vivo (Tran et al., 2007). Automated image analysis was used to identify active compounds, which were subsequently shown to inhibit human endothelial cell tube formation in vitro. Many additional screens can be envisioned as more transgenic models expressing fluorescent markers of organ morphogenesis and physiology are being developed in zebrafish. Crossing these fluorescent reporter lines to the transparent *casper* line will extend the range of detecting fluorescently tagged organs into the adult fish.

Suppression Screens in Zebrafish Mutants

Over the last few years, several zebrafish mutants have been isolated that mimic human disease such as polycystic kidney disease, heart disease, and anaemias (Shin and Fishman, 2002; Zon and Peterson, 2005; Lieschke and Currie, 2007). This offers the possibility to screen chemical libraries for small molecules that confer therapeutic effects in zebrafish mutants. Such chemical suppressors

of a disease process in a whole, vertebrate organism would represent potential lead compounds for drug development. The feasibility of this approach was first demonstrated by the identification of a novel class of compounds capable of suppressing the *gridlock* mutation (Peterson et al., 2004). *Gridlock* mutants have a hypomorphic mutation in the *hey2* gene, a hairy/enhancer of split-related basic helix-loop-helix transcription factor, leading to the malformation of the lateral dorsal aorta preventing blood circulation to the trunk and tail (Zhong et al., 2000). Coarctation of the aorta is a common human congenital cardiovascular malformation that is morphologically similar to the *gridlock* mutant (Towbin and McQuinn, 1995). After screening a diverse library of 5,000 drug-like compounds, two structurally related compounds were identified that completely rescued *gridlock* mutants in a dose-dependent manner to develop a normal vasculature without causing additional developmental defects (Peterson et al., 2004). The mechanism of action is unknown, but the more potent compound, GS4012, induced expression of vascular endothelial growth factor (VEGF). In support of this notion, overexpression of VEGF is sufficient to rescue the *gridlock* mutation and, similar to VEGF, GS4012 promotes human endothelial tube formation in vitro (Peterson et al., 2004). GS4898, which is structurally distinct from GS4012, was identified as a second compound capable of suppressing the *gridlock* phenotype in a second unbiased screen of 7,000 uncharacteristic compounds (Hong et al., 2006). The molecular target of GS4898 is unknown, but pharmacological studies suggest that GS4898 acts by partially inhibiting the phosphatidylinositol-3-kinase (PI3K)/Akt kinase branch of VEGF signaling, which is a negative regulator of VEGF signaling (Hong et al., 2006).

In another example of a chemical suppressor screen in zebrafish, Stern and colleagues screened a chemical library using the recessive cell cycle mutant *crash&burn* (*crb*), a homozygous viable mutation of the *bmyb* gene, that has an increased number of mitotic cells as detected by pH3 antibody staining (Stern et al., 2005). Screening of a 16,320-compound li-

brary resulted in the identification of one compound, named persynthamide, which suppressed the phenotype of the *crb* mutant. Because of its ability to suppress a specific cell cycle defect in *crb* mutants without affecting wild-type embryos, persynthamide might be a useful anticancer agent. An interesting feature of this specific suppressor screen was that 8–10 compounds were pooled for embryo treatment to enable faster processing of the chemical library. A matrix pooling strategy was devised to ensure that each compound was represented in two independent pools, which enabled rapid deconvolution of putative suppressors by cross-referencing (Stern et al., 2005; Murphey et al., 2006). Matrix pooling can greatly increase the throughput, but can complicate identification of the active compound and increase the frequency of toxicity in the screen.

Finally, chemical suppressor screens in zebrafish mutants can also be carried out in transgenic mutant animals expressing a fluorescent reporter gene that permits the in vivo monitoring of organ morphology and physiology. For example, the application of this technology to zebrafish mutants with cardiac rhythm disturbances may lead to the discovery of novel lead compounds for the treatment of heart disease in humans as was recently suggested (Burns et al., 2005). It is worth noting, however, that few zebrafish mutants will be suitable for suppressor screens as one typically needs to identify the rare homozygous mutants that escape embryonic and larval lethality (Margolis and Plowman, 2004). Furthermore, there is the issue of obtaining large numbers of zebrafish with a mutant genotype to screen large, chemically diverse libraries. Taken together, chemical suppressor screens promise to provide a rapid, alternative approach to identify lead compounds for diseases whose underlying biology and pathological mechanisms are not well understood. Further therapeutic development to optimize for specificity will, however, benefit from identification of the molecular targets and understanding of the mode of action of the recovered lead compounds.

Considerations Regarding Compound Concentration, Penetration, and Metabolism

Three key variables in chemical screens that need to be taken into consideration when screening chemical libraries on whole vertebrate organisms are compound concentration, penetration, and metabolism. The concentration of a compound for waterborne treatment of embryos has typically to be increased by an order of magnitude greater than the effective concentrations required for cell culture experiments, where the compounds are directly accessible to all cells. Ideally, each compound of a chemical library would be screened at a full range of concentrations. However, this is impossible in the context of large-scale screens and, therefore, primary screens are usually performed at a single compound concentration. Regarding the choice of an optimal compound concentration, dose-response studies of tyrosine kinase inhibitors provide some guidance. Experiments using the VEGF inhibitors SU5416 and PTK787/ZK222584 demonstrated a complete block of angiogenesis in zebrafish at concentrations of 2 and 5 μM , respectively (Serbedzija et al., 1999; Chan et al., 2002). Furthermore, the FGF receptor inhibitor SU5402 has been extensively used to block FGF signaling in zebrafish and *Xenopus* embryos with doses ranging from 20–160 μM (Raible and Brand, 2001; Shinya et al., 2001; Maroon et al., 2002; Delaune et al., 2005). A survey of the concentrations used to date for chemical screening of zebrafish and *Xenopus* reveals that they ranged between 1 and 100 $\mu\text{g/ml}$ or 5 and 60 μM (Table 2). Overall, an average screening concentration of 10 $\mu\text{g/ml}$ corresponding to 20 μM for small molecules with molecular weights of 500 Da seems to constitute a good compromise in balancing low toxicity yet ensuring that many small molecules reach an active dose in vivo. This notion is also supported by several studies using zebrafish and *Xenopus*, where toxicities associated with higher compound concentrations were reported (Serbedzija et al., 1999; Tomlinson et al., 2009b).

Waterborne exposure of zebrafish or *Xenopus* embryos to small molecules

typically leads to compound uptake initially via the skin and gills, and at later stages by oral adsorption in the gastrointestinal tract through water ingestion. The absorption and bioavailability of a compound in these lower vertebrate animal models cannot be predicted due to the absence of specific data, but will be dependent on the compound's specific physicochemical properties. As in mammalian systems, the compound's molecular weight, hydrophobicity, and number of hydrogen bond donors and acceptors are considered to be critical factors (Lipinski et al., 2001). One parameter, the logarithm of the partition ratio between octanol and water (logP), can be measured empirically or calculated on the basis of the compound's chemical structure. The logP values correlate well with a compound's membrane permeability. A comparison of the logP values of 23 drugs, which were previously tested for bioactivity in zebrafish embryos (Milan et al., 2003), demonstrated that compounds with logP values higher than +1 were absorbed in the zebrafish (Peterson and Fishman, 2004). These findings were confirmed in a subsequent study of 175 bioactive compounds where the logP values of the majority of compounds clustered between +1 and +15 with a marked absence of hydrophilic compounds (Sachidanandan et al., 2008). In terms of the size distribution, the active compounds exhibited no bias and included the full-size range of the library from molecular weights of 200 to 700 Da. The penetration problem of hydrophilic compounds with logP values below 1 can be largely eliminated by microinjection of the compound into the cytoplasm of fertilized eggs or the blood circulation of larvae (Milan et al., 2003). However, these approaches are not suited for large-scale chemical library screening. The drug penetration problem associated with waterborne drug administration can also be seen as an advantage of the screening approach as it allows for the selection of bioactive compounds with favorable permeability or uptake properties in whole organisms.

Drug metabolism is an important factor in the conservation of drug activity across species. The cytochrome P450 (CYP) gene family is a well-rec-

ognized participant in detoxification and drug metabolism. These enzymes not only metabolize a large number of toxic and endogenous compounds, but also participate in biotransformation of ingested drugs. The cytochrome P450 gene family is, therefore, particularly relevant to clinical and pharmacological studies in humans (Danielson, 2002). Metabolism of exogenous compounds in zebrafish and *Xenopus* by the cytochrome P450 (CYP) family enzymes in the liver will undoubtedly affect the compound's stability and may produce a number of bioactive metabolites. The sequencing of mammalian genomes has revealed that the cytochrome P450 gene family is subject to rapid evolution in vertebrate genomes (Gibbs et al., 2004). Compared with human genes, there are clear expansions of several rodent P450 subfamilies, but there are also significant differences between rat and mouse subfamilies. A comprehensive survey of the cytochrome P450 gene content in the zebrafish and *Xenopus* genomes and their phylogenetic relationship with the human cytochrome P450 genes is still lacking and, therefore, it cannot be predicted how exogenous compounds would be metabolized in zebrafish or *Xenopus*. Furthermore, the metabolic activities of the developing livers in zebrafish larvae and *Xenopus* tadpoles may differ significantly from those in adult organisms. Overall, this suggests that species-specific and developmental differences in drug metabolism have to be taken into consideration, when extrapolating data from animal models, even from rodents, to humans.

Despite these limitations, in vivo chemical screens in whole organisms can identify lead compounds that could not have been recovered in in vitro cell-based screens. For example, a screen for compounds affecting erythropoiesis in zebrafish resulted in the identification of a compound that needs to be metabolized in vivo to become effective (Shafizadeh et al., 2004). While the question whether drugs found in zebrafish screens will have similar effects in humans cannot be answered in a general sense, it has been demonstrated that drugs with known effects in humans can cause analogous effects in zebrafish. Milan

et al. (2003) tested 23 compounds in zebrafish known to alter heart rates in humans, which can lead to fatal arrhythmia and constitutes an undesirable drug side effect. Of the 23 drugs tested, 22 caused an analogous prolongation of the cardiac cycle (bradycardia) in zebrafish. In a separate study, the examination of 17 known cell cycle inhibitors in zebrafish embryos revealed that the majority of the tested drugs (9 out of 17) exhibited the predicted cell cycle effects in vivo (Murphy et al., 2006). A further six drugs were shown to be active in a zebrafish cell line but not in embryos, suggesting that they are poorly absorbed into embryos by waterborne treatment. Only two drugs in zebrafish were neither active in vivo nor in vitro indicating their targets are not conserved between zebrafish and mammals. Other drugs that have similar effects in mammals and fish include compounds disrupting angiogenesis by blocking VEGF receptor signaling (Chan et al., 2002). Collectively, these studies suggest that drug targets are generally well conserved between zebrafish and humans, and, therefore, lead compounds identified in zebrafish-based chemical screens are likely to have similar activities in humans.

Approaches to Elucidate Molecular Targets of Bioactive Compounds

While phenotypic whole-organism-based screens have the ability to explore the chemical space for novel compounds with unique biological activities in vivo, the identification of the molecular targets of compounds causing specific phenotypes is not straightforward and offers a major challenge. The identification of the molecular target and mechanism of action is, however, a prerequisite to convert the phenotype-inducing activity of a compound into a molecular understanding of the affected biological process or disease pathway. One approach is to use affinity purification to identify biochemically the specific protein targets of a biologically active small molecule. These experiments are unbiased and do not require assumptions about a molecule's mechanism of action. Strategies on how to perform affinity purifications have

been published elsewhere in detail (Peterson and Fishman, 2004). There are, however, two drawbacks to affinity purification. First, hits recovered from chemical library screens are often moderately potent with a low micromolar affinity (Burdine and Kodadek, 2004). This can lead to non-specific interactions rendering the identification of the primary binding partners difficult. The use of high-affinity compounds would, therefore, increase the success rate of affinity-purification approaches. Secondly, the conjugation of a bioactive compound to a solid resin to enable affinity purification of a target protein may result in the loss of the compound's binding activity. It is, therefore, not surprising that this time-consuming and laborious approach has led to failures in target identification. One alternative approach to overcome the problems is the construction of tagged chemical libraries, where each compound of the library possesses the same functionalized linker that can be used for efficient coupling to solid supports (Ahn and Chang, 2007). Khersonsky and colleagues generated a tagged triazine combinatorial library consisting of 1,536 compounds allowing affinity purification of the target protein and identification by mass spectroscopy (Khersonsky et al., 2003). The tagged triazine library was screened in vivo to recover compounds causing brain, eye, or pigmentation defects in zebrafish embryos (Khersonsky et al., 2003; Jung et al., 2005). One phenotypic screen led to the isolation of a compound that suppressed development of eyes and brain. The tag-free compound named encephalazine was even more potent, suggesting that the tag was not necessary for its activity. Affinity purification led to the identification of four ribosomal subunit proteins (S5, S13, S18, and L28), which were previously reported to be involved in brain and eye development by a genetic mutation (Khersonsky et al., 2003). Other screens using tagged triazine libraries focused on identifying novel pigmentation modulators in cultured melanocytes or zebrafish embryos and have been recently reviewed (Ni-Komatsu and Orlov, 2007). It should be noted that even high-affinity compounds could bind to proteins that are not responsible for

producing the in vivo phenotype of interest. Therefore, it is important to confirm the identity of candidate proteins by performing gene knockdowns using antisense morpholino oligonucleotides. The example given above demonstrates the power of tagged libraries in identifying drug targets without the need for cumbersome structure-activity-relationship studies. It is hoped that the use of tagged libraries will facilitate the process of moving from compound discovery to target identification.

The Screening of Annotated Chemical Libraries Provides Novel Mechanistic Insights

The selection of an adequate chemical library is a crucial step in the process of performing chemical screens. A compilation of the chemical libraries used to date for screens in zebrafish and *Xenopus* is given in Table 2. Two main types of chemical libraries, synthetic combinatorial libraries and collected libraries, can be distinguished. Synthetic combinatorial libraries consist of compounds that are generally all synthesized in parallel through a series of synthetic reactions that join small numbers of building blocks in various combinations to generate large collections of distinct molecules (Schreiber, 2000). Examples of combinatorial libraries used for chemical screens in zebrafish include trisubstituted triazine libraries (Moon et al., 2002; Khersonsky et al., 2003), and libraries containing 1,3-dioxanes (Sternson et al., 2001; Wong et al., 2004) and biaryl-containing medium rings (Spring et al., 2002). Despite the large numbers of distinct compounds contained in these libraries, many of the compounds are structurally related and share common core structures. Hence, they cover only a relatively small area of the chemical space even if they contain thousands of distinct molecules. In contrast, collected libraries are assembled from natural products and/or synthetic compounds that were selected on the basis of chemical diversity, proven biological activity, lack of nonspecific toxicity, and physicochemical properties that are compatible with efficient absorption and bioavailability (Ghose et al., 1999; Lipinski, 2000; Lipinski et al.,

2001; Stockwell, 2004). Generally, the collected chemical libraries frequently possess a larger structural diversity than combinatorial libraries. Consistent with this notion, the hit rates in broad chemical screens for developmental defects are low and comparable for both types of libraries, but fewer distinct phenotypes are observed with combinatorial libraries (Peterson and Fishman, 2004). A further advantage of collected chemical libraries is that they can be obtained from various commercial and non-profit sources as shown in Table 2 and listed in greater detail elsewhere (Peterson and Fishman, 2004). As mentioned above, matrix pooling greatly increases the throughput of chemical library screens (Stern et al., 2005; Murphey et al., 2006). The strategy is, however, primarily suitable for synthetic, combinatorial libraries as many compounds will have no biological activity. In contrast, pooling of compounds from collected bioactive libraries is undesirable as this leads to a very high rate of toxicity (Murphey and Zon, 2006). The choice to pool compounds will, therefore, have to be determined empirically in pilot studies.

Irrespective of the type of library chosen for chemical screening, significant efforts are required to illuminate the mechanistic basis of each compound's activity. To address this problem, Stockwell and colleagues developed a powerful strategy to facilitate the screening step and the study of the lead compound's mode of action (Root et al., 2003). They set out to assemble a chemical library composed of 2,036 well-studied organic compounds with diverse, experimentally confirmed biological mechanisms and activities. Each compound was assigned to one of 169 broad biological descriptors and all published information on its activity was compiled. The library included approved drugs, failed drug candidates, and chemicals with some known biological activity such as antibiotic, anticancer, antidiabetic, and neurological activities among others. The resulting annotated chemical library was structurally more diverse and more enriched in active compounds than conventional, commercially available libraries (Root et al., 2003). Annotated chemical libraries

are a central resource for forward chemical genetics approaches to study biological systems *in vitro* and *in vivo* as was postulated by Stockwell (2000). In recent years, annotated chemical libraries have become a valuable resource for both chemists and biologists interested in carrying out chemical genetic screens. As a consequence, non-profit organizations and companies have developed different annotated chemical libraries. They include the NINDS Custom Collection (National Institute of Health/National Institute Neurological Disease and Stroke), the LOPAC1280 library (Sigma-Aldrich), the Spectrum Collection (Micro-Source), and the ICCB Known Bioactives library (Biomol). These libraries have also been recently used for chemical screens using zebrafish embryos (Table 2) and the outcome of these studies will now be reviewed in greater detail.

The first forward chemical genetics screen was published by North and colleagues in 2007 and was aimed at identifying new pathways modulating definitive hematopoietic stem cell formation during zebrafish embryogenesis (North et al., 2007). They used *in situ* hybridization to monitor changes in hematopoietic gene expression to identify compounds inducing alterations in zebrafish embryos. Most of the non-toxic 2,357 compounds screened were inactive, whereas 35 (1.4%) and 47 (1.9%) increased or reduced hematopoietic stem cell gene expression. Among these substances, 10 affected the prostaglandin pathway. Compounds that increased prostaglandin synthesis promoted hematopoietic stem cell numbers, whereas those that block prostaglandin synthesis reduced the stem cell numbers. The fact that the chemical screen recovered several different agonists and antagonists of prostaglandin synthesis corroborates the findings. Furthermore, the effects of prostaglandins on hematopoietic stem cells were extended into mouse models.

A chemical genetics approach to define the pathways governing vertebrate regeneration was developed using a zebrafish early life stage fin regeneration model (Mathew et al., 2007). The authors screened an annotated chemical library of 2,000 biologically active small molecules in

wild-type zebrafish embryos that had been subjected to caudal fin amputation. They identified 17 compounds including five glucocorticoids, which specifically inhibited regeneration. All functional studies demonstrating a role of glucocorticoids in limiting the regenerative capacity of the fin were performed using a prototype glucocorticoid receptor agonist. Since the identity of the biologically active compounds was not disclosed, the study is of limited value to researchers active in the field of tissue regeneration. By contrast, all hits were documented in a chemical genetics screen for antiangiogenic activities in transgenic zebrafish (Tran et al., 2007). Screening of the LOPAC1280 library resulted in the identification of two known antiangiogenic compounds and one compound, indirubin-3'-monoxime (IRO), not previously known to possess antiangiogenic activity. IRO was subsequently shown to inhibit human endothelial tube formation *in vitro*. IRO is a cell-permeable derivative of indirubin and was shown to inhibit a number of kinases including Lck, cyclin-dependent kinases, and AMP-activated protein kinases. Therefore, the antiangiogenic activity of IRO may be multimodal resulting from the inhibition of multiple kinases (Tran et al., 2007).

Another example demonstrating the power of screening annotated chemical libraries in whole organisms led to the identification of dorsomorphin, the first small molecule inhibitor of the bone morphogenetic protein (BMP) signaling pathway (Yu et al., 2008b). The authors decided to use an *in vivo* screening approach that would identify compounds that inhibit BMP signaling while selecting against those with non-specific biological effects. Since gene mutations that disrupt BMP signaling affect dorsoventral patterning resulting in dorsalization, the authors screened several small molecule libraries for compounds that would phenocopy these mutants. The chemical libraries screened totaled over 7,500 compounds and contained among others known bioactive molecules and approved drugs. However, only dorsomorphin produced substantial and reproducible dorsalization in zebrafish embryos. In a series of *in vitro*

studies, the authors show that dorsomorphin inhibits phosphorylation of the receptor SMAD proteins, probably by blocking type I BMP receptors. Interestingly, dorsomorphin was previously shown to antagonize AMP-activated kinase (AMPK) *in vitro* (Zhou et al., 2001), an enzyme that has no role in dorsoventral patterning in zebrafish as demonstrated by Yu and colleagues (Yu et al., 2008a). Dorsomorphin was later used as a valuable novel tool to probe the requirement for BMP signaling in osteoblast differentiation *in vitro* and bone mineralization *in vivo*, and to explore the physiological role of hepatic BMP signaling in iron metabolism (Yu et al., 2008a). On the basis of the latter findings, dorsomorphin has been suggested as a promising lead compound for the treatment of anemia in chronic disease. Furthermore, preclinical studies of dorsomorphin and its subsequently developed more potent and specific derivatives (Cuny et al., 2008) have also suggested a rational therapy to treat fibrodysplasia ossificans progressiva (FOP), a congenital disorder of progressive and widespread postnatal ossification of soft tissues, and other heterotopic ossification syndromes associated with excessive BMP signaling (Yu et al., 2008a).

Finally, a phenotype-based chemical library screen of 5,760 compounds for novel regulators of zebrafish embryogenesis resulted in the identification of DTAB, a novel retinoid that selectively activates the retinoic acid receptor γ (RAR γ) and disrupts patterning along the anterioposterior body axis *in vivo* (Sachidanandan et al., 2008). Overall, these studies demonstrate that the screening of annotated chemical libraries in whole organisms can yield novel regulators of biological processes such as mitosis, regeneration, stem cell proliferation, or organogenesis, and can identify novel, previously elusive inhibitors of important signaling pathways. Furthermore, they highlight the great advantages that the novel compounds emerging from these *in vivo* screens provide with their exquisite temporal control in the dissection of signaling pathways and their promising potential as novel lead compounds for drug development.

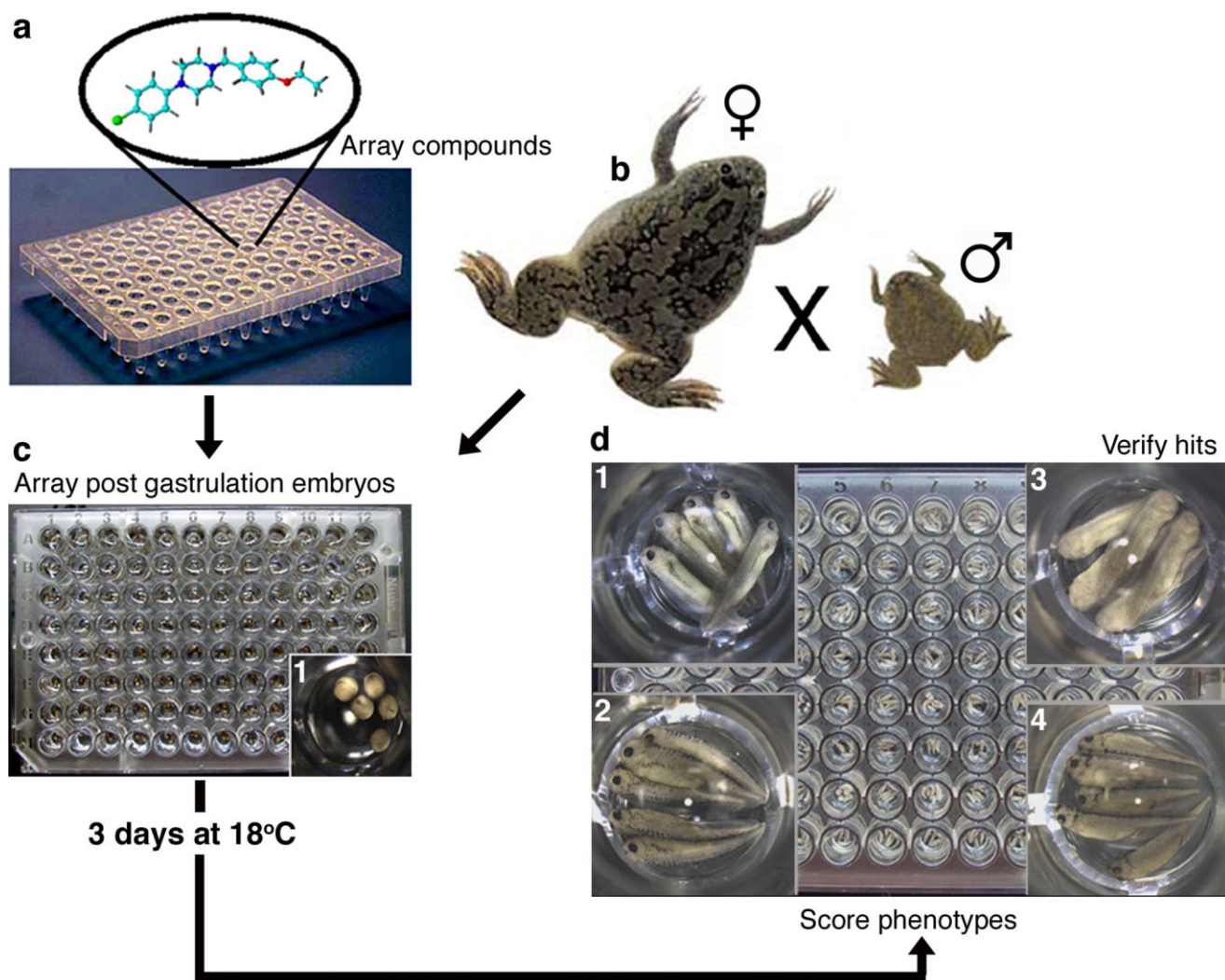


Fig. 2. Flow diagram of a *Xenopus* chemical library screen. Compounds are arrayed in multi-well plates and five embryos at stage 15 are added per well. These are incubated for 3 days and then scored for phenotypic changes in comparison to control tadpoles. Examples of some of the possible pigmentation phenotypes observed are shown. Positive hits are independently verified. Reproduced by permission of Royal Society of Chemistry (Tomlinson et al., 2009b).

XENOPUS AS A NEW MODEL FOR LARGE-SCALE CHEMICAL SCREENS

A General Outline for Chemical Screens in *Xenopus*

Despite the fact that *Xenopus* make an ideal tetrapod model for in vivo chemical screens due to the small size of the embryos, accessibility during development ex utero, optical transparency of tadpoles, permeability to small molecules, and favorable evolutionary position relative to humans, their utility has remained untested until recently. The feasibility of performing large-scale phenotype-based

chemical screens in *Xenopus* is illustrated by two recent examples in which novel compounds affecting either pigmentation or angiogenesis and lymphangiogenesis were identified (Tomlinson et al., 2009a,b; Kälén et al., 2009).

The general procedure of performing large-scale chemical screens with *Xenopus* embryos is outlined in the flow diagram shown in Figure 2. Typically, compound treatment is not initiated before the completion of gastrulation, unless the screen is targeted for biological processes occurring during blastula stages or gastrulation. Pilot studies have shown the suggested timing of chemical exposure is benefi-

cial as the toxicity of compounds is much less pronounced in older embryos (Tomlinson et al., 2005) (Kälén and Brändli, unpublished observations). Unfertilized eggs and malformed embryos can be removed, whose death would otherwise contribute to poor health, developmental delays, and often death of the remaining embryos in the well. In addition, it is easier to select healthy, developmentally synchronized embryos for chemical treatments at later stages. Embryos or tadpole, usually five per well, are arrayed in 96- or 48-well microtiter plates, respectively, containing the culture medium supplemented with the test compounds at the desired con-

centration. Typically, a low-magnification dissection microscope is used to monitor the embryos for phenotypic changes during cultivation in presence of the compounds. The examples of embryos displaying differences in pigment patterns, which are easily detected by visual inspection, are shown in Figure 2. Other rapidly detected phenotypes occurring in response to compound treatment include defects in brain and eye morphogenesis (e.g., small eyes), and effects on heart function and the shape of the trunk, tail, and fins.

A Chemical Screen for Compounds Affecting Pigment Cell Development

Pigment cells (melanophores or melanocytes as they are more commonly known in higher vertebrates) are a particularly useful cell type to focus on for high-throughput chemical screening. This is due to the ease of visual scoring and because melanophores and retinal pigment epithelial (RPE) cells are capable of acting as an *in vivo* model for many aspects of cellular biology, including cell migration, morphology, and biochemical pathways (such as the production of pigment). In addition, the development of pigmented tissue is related to a number of disease states and so a screen focusing on melanocytes could also play a role at the early stages of drug discovery. Examples of such diseases include albinism, piebaldism, an autosomal disorder leading to localized hypopigmentation (Thomas et al., 2004), vitiligo, an autoimmune disease leading to the destruction of melanocytes and pigment-free patches of skin in patients (Kelsh et al., 2000), hyperpigmentation, and finally skin cancer or melanoma (Kelsh et al., 2000; van Kempen and Coussens, 2002; White and Zon, 2008; Sturm, 2009).

We have carried out a developmental chemical genetic screen of 2,950 compounds to identify phenotypes associated with pigment cell development in *Xenopus* (Tomlinson et al., 2009b). We recovered 41 compounds producing various phenotypes (Tomlinson et al., 2009b). Two compounds showed a reduction in the size of the eye and several compounds caused general developmental defects and edema in the

heart and kidneys. The largest number of compounds discovered in one phenotypic category were those causing a partial or total loss of pigment in the melanophores and retinal pigment epithelium (17 compounds). Five compounds were identified as affecting melanophore morphology in the developing embryo. Seven compounds were identified that affected the normal migration of melanophores. Of these, we have further characterized compound NSC 84093, which gives a striking vertical banding pattern on the dorsal side of the embryo and a decrease in the melanophores on the ventral side of the tail. We identified this compound as an 8-quinolol derivative and further demonstrated that it functions as a potent matrix metalloproteinase (MMP) inhibitor. MMPs are a family of proteins known to have important roles in cell migration, inflammation, angiogenesis, and cancer (Page-McCaw et al., 2007). Potential targets for NSC 84093 included MMP-14 and MMP-2, which are expressed in or close to migrating neural crest cells and will give rise to melanophores. Interestingly, knockdown of MMPs using morpholinos partially phenocopied the effect of NSC 84093 (Tomlinson et al., 2009a). NSC 84093, therefore, represents a novel and unique molecular tool to investigate neural crest cell migration and subsequent melanophore development *in vivo*. Furthermore, NSC 84093 may contribute towards a better understanding of the roles of MMPs in cell migration processes under normal and pathological conditions, such as in cancer.

A Chemical Screen for Compounds Affecting Angiogenesis and Lymphangiogenesis

Angiogenesis and lymphangiogenesis are essential for embryonic development and organogenesis, but also play important roles in tissue regeneration, chronic inflammation, and tumor progression (Carmeliet, 2003; Alitalo et al., 2005; Cueni and Detmar, 2006). Despite the significant advances in developing antiangiogenic therapies of recent years, the identification of novel drugs targeting the blood and lymph vasculature is of high priority. This is particularly true for lymphangiogenesis, where the lack of an appropriate, simple ani-

mal model has hampered progress in elucidating the key molecular players and signaling pathways. In addition, large-scale *in vivo* screens to identify drug-like small molecule modulators of lymphatic vessel formation are missing to date. The recent imaging of an elaborate lymph vessel system and its molecular characterization in *Xenopus* tadpoles (Ny et al., 2005) has created new opportunities for studying vascular development and chemical screens.

We have applied an unbiased forward chemical genetics approach in combination with a simple phenotypic readout to uncover signaling pathways and small molecule regulators of lymphatic and blood vascular system development in *Xenopus* tadpoles (Kälin et al., 2009). In a secondary screen, *Xenopus* embryos were treated with the hits recovered from the primary screen and analyzed by *in situ* hybridization for defects in angiogenesis and/or lymphangiogenesis. An annotated chemical library (LOPAC1280, Sigma-Aldrich) representing small organic molecules with diverse well-defined pathway specificities was analyzed using this novel two-step chemical screening strategy. After screening 1,280 small molecules, 65 compounds scored positive in the primary screen and were subsequently subjected to the secondary screen. This resulted in the identification of 32 compounds modulating distinct aspects of vascular development *in vivo*. Specifically, we found that 18 compounds selectively blocked blood vessel development, six compounds interfered with both blood and lymphatic vessel development, and eight compounds affected lymphatic development only. The bioactive compounds fell into 15 distinct pharmacological classes with modulators of phosphorylation representing the largest activity class (13 out of 32). Positive hits included inhibitors of the VEGF signaling pathway, which is essential for vascular development (Olsson et al., 2006), and thus validates the *in vivo* chemical screening approach. This point is further underscored by the fact that the chemical screen in *Xenopus* was able to recover five of the eight known antiangiogenic compounds represented in the LOPAC1280 library. By contrast, a screen of the LOPAC1280

library for antiangiogenic activity in transgenic zebrafish resulted in the recovery of only three out of the eight antiangiogenic compounds (Tran et al., 2007). This indicates that the chemical library screening strategy in *Xenopus* is more sensitive in recovering compounds with vascular activity than the one employing transgenic zebrafish.

The chemical screen in *Xenopus* also resulted in the identification of small-molecule modulators of pathways not previously known to mediate vascular and/or lymphatic development. These included, for example, an adenosine A1 receptor antagonist that inhibited both lymphatic and blood vessel formation in *Xenopus* tadpoles. In a proof-of-principle study, we subsequently validated this compound in a mammalian animal model by demonstrating that it blocked VEGFA-induced neovascularization in adult mice. Taken together, the study established a rapid and sensitive in vivo two-step method for large-scale chemical screens using *Xenopus* tadpoles to identify novel pathways and lead compounds with selectivity for lymphatic and blood vessel formation in a time- and cost-saving manner. The recovered compounds represent a rich resource for in-depth analysis in the future, and their drug-like features will facilitate further evaluation in pre-clinical models of inflammation and cancer metastasis.

Inverse Drug Screening

Whereas large-scale chemical library screens have until recently not been carried out in *Xenopus*, the use of pharmacological agents to study developmental processes has a longer tradition. Defined small-molecule antagonists are valuable tools in pharmacological loss-of-function studies in vivo. They can be used at any time of development and are ideal to study the role of maternal proteins, which is otherwise difficult to achieve using genetic or knockdown approaches. The use of individual chemical inhibitors as tools for functional analysis of *Xenopus* development has, for example, been demonstrated using cyclopamine, an inhibitor of hedgehog signaling, SU5402, a potent fibroblast growth factor receptor (FGFR) inhibitor, and the retinoic acid inhibitor

BMS453 (Koebernick et al., 2003; Chen et al., 2004; Chung et al., 2004). On a larger scale, the effects of treating *Xenopus* embryos with an array of 13 protein kinase inhibitors demonstrated that several compounds caused specific developmental defects (Tomlinson et al., 2005). In general, these approaches are frequently limited by the lack of suitable, highly specific drugs that either inhibit or activate a given protein function or signaling pathway in vivo. For example, to date there are no reliable small-molecule inhibitors of the Wnt signaling pathway known (Barker and Clevers, 2006). By contrast, rich collections of pharmacologic agents are available for certain protein families, such as ion channels, neurotransmitter receptors, and pumps, which could be tapped to implicate specific protein families in a chosen biological process.

Levin and co-workers took advantage of this resource and have developed a chemical genetic screening strategy coupled with a simple phenotypic read-out to rapidly gain molecular insight into endogenous ion flows that control the establishment of left-right asymmetries of the body plan in *Xenopus* embryos (Levin et al., 2002). In their screening strategy, fertilized eggs were treated with drugs of increasing specificity in successive iterations of the screen to narrow down the lead candidate protein(s). Using 14 broad-range inhibitors, most protein families tested, such as sodium and chloride channels, were ruled out to play a role in left-right asymmetry. However, the broad-range screen implicated that blocking H^+ and K^+ ion flux induced heterotaxia, an abnormal arrangement of organs or parts of the body in relation to one another. The application of more specific inhibitors identified the H^+/K^+ -ATPase as the ion pump responsible for establishing differential hydrogen/potassium fluxes as an early step in the determination of left-right asymmetry in *Xenopus* embryos. This conclusion was then supported by genetic disruption of endogenous H^+/K^+ -ATPase activity, and by pharmacological studies in chicken embryos (Levin et al., 2002).

Adams and Levin subsequently formalized their iterative screening strategy, which they termed inverse

drug screening (Adams and Levin, 2006). The pharmacological "loss-of-function" strategy uses known chemical agents to rapidly implicate specific candidates for roles in any chosen biological process. The hierarchical testing procedure allows the assessment of large numbers of pharmacological agents in a manner that is more efficient than performing exhaustive screens of entire compound families. No more than 20–30 compounds need to be tested before implicating a short list of targets that are ultimately validated. Hence, the strategy quickly reveals a manageable number of specific molecular candidates that can then be targeted and validated in detail using highly specific pharmacological agents and genetic approaches, respectively. To date, inverse drug screening in *Xenopus* embryos has been used to implicate H^+ -V-ATPase-dependent proton fluxes in early left-right patterning and tail regeneration (Adams et al., 2006, 2007) and to uncover novel pre-nervous roles for the neurotransmitter serotonin in left-right patterning (Fukumoto et al., 2005a,b). Inverse drug screening, therefore, represents an interesting, alternative chemical genetic approach to dissect biological process using small-molecule effectors.

Using Cell-Free *Xenopus* Extracts for Chemical Screens

While the focus of this review is on whole-organism-based chemical screens, *Xenopus* has also been instrumental in the development of chemical screening methods using cell-free extracts. Unique to *Xenopus* among model organisms is the use of oocyte and egg extracts as cell-free systems for the study of various cellular and biochemical mechanisms (Liu, 2006). *Xenopus* oocytes and eggs contain the complete biochemical machineries necessary for cell cycle progression and DNA replication. Cell-free *Xenopus* extracts can be obtained in large quantities, are easily activated in vitro, and can recapitulate important cellular processes such as cell cycle progression, DNA replication, and centrosome duplication in vitro. Given these properties, it is not surprising that cell-free extracts from *Xenopus*

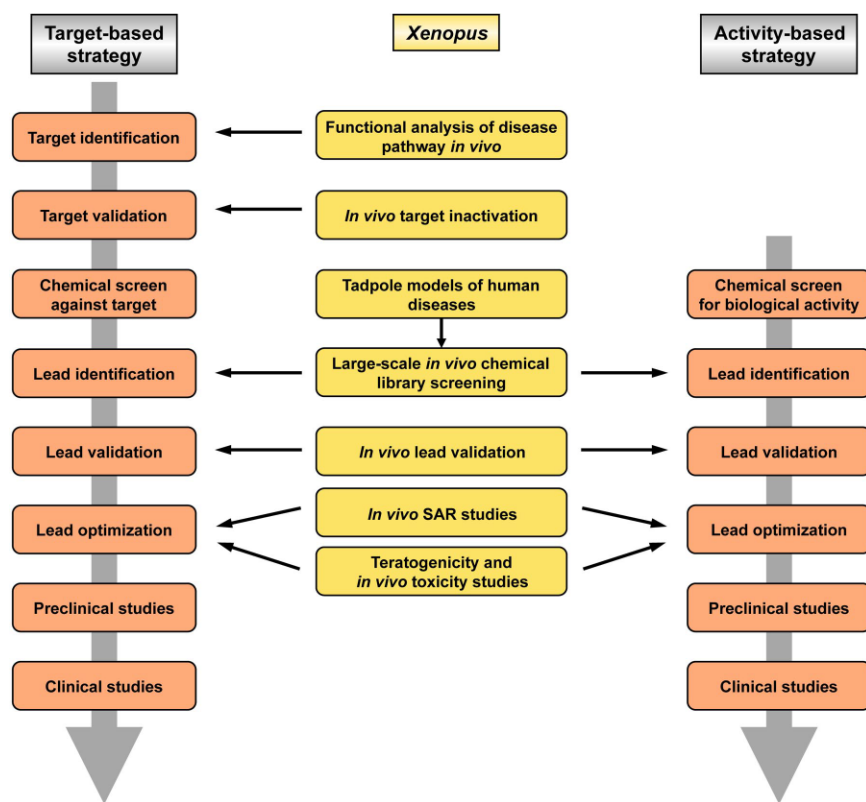


Fig. 3. The utility of *Xenopus* in drug discovery and development. The two broad strategies of drug discovery and development are shown. The target-based strategy requires a functional understanding of the disease pathway to determine target genes or proteins for drug discovery and development. The activity-based strategy selects for drug candidates on the basis of biological activities implicated with a therapeutic benefit. The *Xenopus* embryos and tadpoles can serve as cheap and efficient bioassay tools at several stages of the drug-discovery process regardless of the strategy chosen. SAR, structure-activity-relationship.

have been used successfully in chemical screens to identify small-molecule modulators of actin assembly, microtubule stability, ubiquitin-dependent protein degradation, cell cycle machinery, and DNA damage response (Verma et al., 2004; Wignall et al., 2004; Peterson et al., 2006; Dupre et al., 2008; Landais et al., 2009). Chemical screening in cell-free *Xenopus* egg extracts takes place in a fully soluble *in vitro* context. Therefore, compound delivery is simple and problems associated with compound toxicity and bioavailability are minimized. The utility of the compounds emerging from these screens for the treatment of cells or organisms needs to be established in each case. Despite this limitation, chemical screens in *Xenopus* extracts have the potential to reveal novel drugable components in important biochemical pathways (Hathaway and King, 2005).

***XENOPUS* AS A SCREENING TOOL IN THE DRUG DISCOVERY AND DEVELOPMENT PROCESS**

Basic Strategies in Drug Discovery and Development

There are two broad strategies for drug development, which converge at the level of the lead validation step and are followed by preclinical and clinical studies (Fig. 3). The target-based strategy has been applied in drug discovery and molecular pharmacology over the past decade. It requires a functional understanding of the disease pathway to determine specific target proteins or genes directly associated with the disease. Suspected targets are subjected to validation to determine whether they are critical to disease pathogenesis. Con-

firmed targets will serve as a basis for large-scale high-throughput *in vitro* screens to identify active compounds. The cell-free assays typically monitor the ability of a compound to bind to the target and/or inhibit its function. The *in vitro* approach has the advantage that target–compound interactions are identified in the absence of confounding variables. Active compounds recovered from the screening of chemical libraries will be used as leads that will have to undergo validation and optimization before they or their optimized chemical derivatives can enter preclinical and clinical studies. In the activity-based strategy, the molecular basis of a disease is unknown and active compounds cannot be identified on the basis of their inhibition of a specific protein target. Therefore, large-scale chemical screening is performed against an intact biological system to identify lead compounds that induce interesting phenotypic changes. Activity-based approaches are also known as phenotype-based drug discovery approaches. The biological systems suitable for phenotypic chemical screening can consist of prokaryotic or eukaryotic single-cell organisms, normal or pathological mammalian cells, or even whole multicellular organisms. Once biologically active compounds that induce a desired phenotype are identified, they can enter the lead optimization and development process. In addition, efforts are directed at studying the mechanistic basis of the phenotype by identifying the target. The structural and biological information of the target can be used in target-based approaches to identify and develop more potent compounds that disrupt the function of the target. The end point of any drug discovery and development process is a drug, a registered and approved chemical substance that is used in a dose-dependent manner in treatment of a medical condition.

Identifying Drug Targets *In Vivo*

Regardless of the strategy chosen, *Xenopus* embryos and tadpoles can serve as cheap and efficient bioassay tools at several stages of the drug discovery

process (Fig. 3). At present, the utility of *Xenopus* for therapeutic target identification is unproven, but promising examples are emerging (Kälin et al., 2007; Tomlinson et al., 2009a). Ample genomic and transcriptomic information and the similarity to mammals make *Xenopus* amenable to functional analysis relevant to uncovering human disease pathways and the identification of targets suitable for drug development. For example, a large-scale *in situ* hybridization analysis of solute carrier (SLC) gene expression has provided compelling evidence that the *Xenopus* orthologues of eleven SLC genes causing inherited renal diseases in humans are expressed in the equivalent segments of the *Xenopus* pronephric kidney (Raciti et al., 2008). The *Xenopus* pronephros can, therefore, serve as a novel model for the study of human renal disease mechanisms. Similarly, the molecular pathways regulating angiogenesis and lymphangiogenesis, two processes that play central roles in wound healing processes and in the pathology of inflammatory diseases and cancer, are shared between *Xenopus* and mammals (Cleaver and Krieg, 1998; Helbling et al., 2000; Ny et al., 2005; Cox et al., 2006; Kälin et al., 2007). Molecular dissection of these pathways can be done efficiently and rapidly by targeted gene knockdowns using antisense morpholino oligonucleotides as demonstrated first for the canonical, β -catenin-dependent Wnt signaling pathway in *Xenopus* (Heasman et al., 2000). Alternatively, the screening of annotated chemical libraries can identify novel targets as demonstrated for matrix metalloproteinases, which were recently shown to play an essential role in pigment cell migration (Tomlinson et al., 2009a). Finally, large-scale morpholino knockdown screens (Kenwrick et al., 2004; Rana et al., 2006), genetic screens by random mutagenesis in *Xenopus tropicalis* (Goda et al., 2006), and novel targeted mutagenesis approaches (see Future Directions section) will make *in vivo* functional analysis of disease pathways and therapeutic target identification in this vertebrate model organism an even more attractive alternative to expensive and time-consuming mouse models.

Validating Drug Targets In Vivo

Target validation encompasses the demonstration that a target gene or protein is critical to a signaling pathway underlying a medical condition or to the pathogenesis of a disease. A rigorous evaluation must occur to demonstrate that modulation of the target will have the desired therapeutic effect. This involves intensive *in vitro*, as well as *in vivo* studies that provide gain- and loss-of-function information on the possible effects of therapeutic intervention. The result of these efforts is to establish sufficient knowledge so that physiologically relevant model systems can be developed into assays for downstream chemical screening. Current validation methods require time-consuming and often expensive studies, particularly if transgenic mouse models have to be developed to validate a target. Target validation has, therefore, become a bottleneck in drug development. On the other hand, the failure to correctly determine the biological relevance of a target can result in large accumulated costs due to high attrition rates on the later stages of drug development. Thus, alternative approaches that streamline and speed up the target prioritization efforts are of immense value to the drug development community. Lower vertebrate animal models such as *Xenopus* offer a cheap alternative to demonstrate the causality of target gene candidates *in vivo*. Gene overexpression and morpholino knockdown experiments in *Xenopus* embryos can be used to validate targets and to identify epistatic relationships between individual targets and other pathway members. For example, the G protein-coupled receptor APJ and its ligand have been implicated in regulating tumor angiogenesis (Kälin et al., 2007). Using gain- and loss-of-function experiments in *Xenopus*, the role of APJ and apelin in regulating embryonic angiogenesis was determined and the epistatic relationship between VEGF and apelin-APJ signaling was established (Cox et al., 2006; Kälin et al., 2007).

Validating Lead Compounds In Vivo

Once a target has been validated, large-scale high-throughput chemical screens are carried out to identify lead

compounds. *In vitro* chemical screening against a defined molecular target exploits a protein's function, i.e., enzymatic activity, signaling activity, or its interaction with its partners, to identify small-molecule binders. Alternatively, activity-based approaches use large-scale *in vivo* chemical screening in whole organisms to identify leads on the basis of phenotypic changes. The utility of *Xenopus* as a novel vertebrate model for *in vivo* chemical screening has recently been established (Tomlinson et al., 2009a,b; Kälin et al., 2009) and is extensively discussed above. Regardless of the primary screening strategies, the identified small-molecule hits are typically confirmed using a series of secondary assays. In this confirmative process known as lead validation, *Xenopus* embryos and tadpoles can be employed as powerful *in vivo* models. This is best documented for drug discovery screens aimed at interfering with the Wnt signaling pathway, where the β -catenin-dependent transactivation of Tcf-dependent genes represents the key molecular lesion causing colorectal cancer (Barker and Clevers, 2006). Various cell-based chemical libraries screens have identified small molecules that modulate β -catenin-dependent Wnt signaling *in vitro*, including the agonists 6-bromoindirubin-3'-oxime (BIO) (Meijer et al., 2003) and a pyrimidine derivative (Liu et al., 2005); the co-activator QS11 (Zhang et al., 2007); the antagonists NSC668036 (Shan et al., 2005), a natural product compound (Lepourcelet et al., 2004); and other synthetic compounds identified using a virtual compound screening strategy (Héligon and Brändli, unpublished observations). In all these cases, *Xenopus* embryos served as effective and rapid *in vivo* validation systems that provide a basis for rational design of high-affinity compounds modulating Wnt signaling *in vivo*. Other promising areas in pharmaceutical research, where the use of *Xenopus* as an *in vivo* bioassay tool for lead validation is expected to accelerate the drug discovery process, include the development of small-molecule inhibitors of (lympho)angiogenesis and skin pigmentation.

Optimizing Lead Compounds In Vivo

Before validated lead compounds enter preclinical studies, they are optimized through structure-activity-relationship (SAR) studies to improve their properties, such as increasing their potency, specificity, and enhancing their pharmacokinetic properties. SAR studies require the synthesis of analogous synthetic compounds with refined properties. Besides increased potency and target selectivity, drug-like characteristics that are compatible with testing in biological model systems are improved. The synthetic analogues are subjected to assay-based screening to compare their physicochemical and biological properties with those of the initial lead compounds. Repeated rounds of SAR studies give rise to progressive lead optimization. SAR studies are typically performed in vitro either by assessing binding to the purified target protein or using cell culture-based assays. Apart from the intrinsic biological activity on the target, the absorption, distribution, metabolism, and excretion (ADME) are fundamental pharmacokinetic properties that determine the in vivo efficacy of a drug candidate. Structural changes that improve the potency of a compound may have detrimental effects on ADME, which go undetected in traditional in vitro SAR studies. By performing SAR studies in whole organisms, structures can be identified that improve potency without loss of in vivo efficacy or increased toxicity. *Xenopus* embryos and tadpoles are highly amenable to in vivo SAR studies as recently demonstrated for NCS 84093, an inhibitor of pigment cell migration, and other structurally related molecules (Tomlinson et al., 2009a). One danger associated with performing in vivo SAR studies in *Xenopus* is that the screening may select for compounds that are species-specific and may no longer recognize the human protein target. Validation of optimized leads in human in vitro assays is, therefore, imperative. However, given the ease of performing the SAR experiments in *Xenopus* embryos and tadpoles, it is likely that in vivo SAR studies will become increasingly popular. Another attractive aspect of per-

forming in vivo SAR studies is that they couple the characterization of binding affinities with ADME analysis. Compound testing in *Xenopus* can facilitate the identification of lead derivatives with improved absorption properties or favorable metabolic conversion rates resulting in enhanced bioavailability and drug efficacy. Given the differences in physiology and metabolism, it is evident that the ADME properties of small molecules in *Xenopus* embryos and tadpoles will not reflect all aspects of ADME in adult mammalian animal models or humans. Nevertheless, structural optimization of leads in *Xenopus* represents a simple, rapid, and cost-effective experimental approach that can assist in the prioritization process of compounds for subsequent testing in more sophisticated mammalian animal models.

Assessing Toxicity and Teratogenicity

Toxicology and teratology testing are other important areas of drug development, where *Xenopus* serves as convenient vertebrate animal model. In fact, *Xenopus* have a long history in the fields of teratology and toxicology. The Frog Embryo Teratogenesis Assay-*Xenopus* (FETAX) is a 96-hr whole-embryo developmental toxicity test that uses late blastula-stage *Xenopus* embryos to measure the effects of chemicals on mortality, malformation, and growth inhibition (Courchesne and Bantle, 1985). FETAX has been used for many years as a screening assay to identify potential human teratogens, toxic compounds, and ecological hazards as demonstrated by two recent examples, where human pharmaceuticals and an antimalarial drug were assessed (Richards and Cole, 2006; Longo et al., 2008). The advantages of FETAX include its speed and low cost. It is, however, not clear how predictive FETAX will be of human toxicity, since *Xenopus* have limited xenobiotic metabolism through the first 96 hr of embryogenesis. These limitations can be partially overcome using an in vitro metabolic activation system, where test compounds are pretreated with activated rat liver microsomes (Fort et al., 1988). Using this pretreatment procedure, the teratogenic risk

of proteratogenic compounds, such as thalidomide and aflatoxins, can be accurately assessed (Fort et al., 2000; Vis-mara and Caloni, 2007). Taken together, *Xenopus* represents a promising bioassay system in the drug discovery and development process that will complement cell culture assays, and in vivo experiments in mice. Ideally, *Xenopus*-based assays are positioned at an early stage in the lead validation and optimization pipeline.

FUTURE DIRECTIONS

Phenotype-based chemical library screens in zebrafish and *Xenopus* embryos have emerged as powerful approaches to recover bioactive small molecules, identify novel therapeutic lead compounds, and characterize organ-specific toxicities of drug candidates in preclinical development. Analogous approaches are not possible in rodents because their development occurs in utero and embryos are limited in number. With the increasing costs of drug development, the identification of cheaper ways to evaluate lead compounds in an in vivo situation will be of interest to pharmaceutical companies. *Xenopus* and zebrafish also offer complementary models to the mouse as they can provide an initial assessment of the ADME properties of a compound under development, which tests in cell cultures are unable to do.

As shown in Table 2, the number of large-scale chemical screens carried out to date in *Xenopus* is fairly limited. There is, therefore, plenty of room for further chemical screens targeting different tissue and organ systems. By varying the time point and length of compound exposure, different aspects of organogenesis, such as specification, patterning, and terminal differentiation, can be targeted, enabling the chemical dissection of the complex biological processes underlying organogenesis. In future, chemical genetic screens will also take advantage of *Xenopus* as a model tetrapod to study organs and appendages not found in fish, such as lungs, lymph hearts, and limbs. This asset and, in comparison to zebrafish larvae, the increased similarities of the tissues and organs of *Xenopus* tadpoles to their mammalian counterparts will trans-

late into a higher predictive power of findings made in *Xenopus* for our understanding of human biology. Although highly informative and powerful, the large-scale chemical genetic approaches used to date remain laborious, especially if the desired phenotypes cannot be scored by simple visual inspections of intact embryos and tadpoles and, therefore, require in situ hybridization or immunohistochemical methods to detect chemically induced changes. Promising automation of the screening process and intelligent compound pooling strategies to increase the through-put of chemical screens have been reported (Burns et al., 2005; Stern et al., 2005; Tran et al., 2007) and recent developments, such the use as transgenic GFP reporter lines (Tran et al., 2007), will enable more sophisticated screens to be carried out in the future. These will include screens targeting internal organs, such as the gut, immune system, and vasculature inaccessible to visual inspection of intact embryos. Furthermore, chemical screens may be devised that use GFP reporters to monitor changes in physiological activities.

A further focus of intense research in the future will be on the development of lower vertebrate models of human disease for whole-organism-based chemical library screening. First examples of chemical screens to identify small molecules that suppress disease phenotypes in mutant zebrafish and thus provide therapeutic benefit to the affected animals have been reported (Peterson et al., 2004; Stern et al., 2005; Hong et al., 2006). The identification of genetic mutants in *X. tropicalis* (Goda et al., 2006) will complement the many mutants already available in zebrafish. Presently, chemical suppressor screens are limited to the pool of mutants that have emerged from various random mutagenesis screens carried out in zebrafish and *Xenopus*. Many of these mutants are not suitable for chemical suppressor screens as they do not have robust or reproducible phenotypes and there is frequently the issue of obtaining sufficient amounts of mutant individuals (Margolis and Plowman, 2004). The development of zebrafish and *Xenopus* models that accurately phenocopy human congen-

ital diseases requires the generation of tailor-made mutations in the orthologues of human disease genes. Targeted knockout approaches and the engineering of genes carrying predetermined mutations has been very successful in mice, but are not feasible in zebrafish and *Xenopus* due to the inability to grow embryonic stem cells. A step forward towards recovering mutations in a gene of interest has been the introduction of TILLING (Targeting Induced Local Lesions IN Genomes) methodology (Wienholds et al., 2003; Stemple, 2004; Moens et al., 2008). This reverse genetics strategy utilizes the screening of chemically mutagenized genomes by resequencing and TILLING to identify mutations in specific genes. In addition to zebrafish, TILLING has recently also been successfully applied to recover recessive mutations in *X. tropicalis* (D.L. Stemple, personal communication). Although specific regions within genes can be analyzed for disruptions, the mutations obtained by TILLING are random. Moreover, the required resources and technical knowhow are beyond the scope of most laboratories. Recently, zinc-finger nuclease technology has been developed that offers the possibility to engineer mutations in specific zebrafish genes (Doyon et al., 2008; Meng et al., 2008).

The technique employs zinc-finger nucleases, chimeric molecules consisting of a DNA binding zinc-finger domain and the FokI restriction endonuclease, to induce targeted mutations in any gene of interest. While not proven to date, the technique offers a rational and straightforward strategy to engineer precise models of human inherited diseases in zebrafish and *X. tropicalis*. Given the diploid nature of the *X. tropicalis* genome and the extensive synteny shared with the human genome, we anticipate that the engineering of animal models carrying mutations in human disease genes will be more successful in *X. tropicalis* than zebrafish, whose genome has undergone an additional genome duplication relative to tetrapods. The modeling of human diseases in *Xenopus* carries the great hope that thousands of mutant embryos can be obtained from a single mating, which will subsequently be employed for

large-scale chemical suppressor screens to identify novel compounds conferring therapeutic activity. While this approach is presently already possible using transgenic mouse models, costs and limitations in obtaining sufficient amounts of mutant animals have prohibited the practical execution of such screens. In our opinion, the modeling of human inherited diseases affecting skin pigmentation, the heart, the kidney, the blood vasculature and lymphatics will be of particular interest as defects in these tissues and organ systems will not cause early embryonic lethality and first become evident once tadpole stages have been reached. Affected tadpoles will display phenotypes that are easily identified by visual inspection, such as loss of pigmentation, edema formation, and irregular heart beating, which offers the advantage that small molecules suppressing the disease pathology can be easily scored and recovered. The coming years will show whether these exciting new possibilities can be exploited. Overall, the development and use of multicellular vertebrate organisms, such as zebrafish and *Xenopus* embryos, as in vivo drug discovery tools has expanded our possibilities to investigate the chemical universe for novel small bioactive molecules that could serve as lead compounds for drug development and therapeutic intervention.

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